Lab 2: IBM Quantum Composer Circuits

Pauli-X Flip

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1. Introduction

This experiment provides a introduction to using IBM quantum circuit composer application by runing a Pauli-X Flip gate circuit with a qubit prepared in the |0⟩ state|. The expected outcome is for the qubit to have a 100% probability of being in the |1⟩ state

2. Methods

Backend Details

Backend	ibm_fez
Job ID: run 1	d6fjls0ddp9c73ceg1kg
Job ID: run 2	d6fjoja24kpc73elq6d0
Job ID: run 3	d6fjq8a24kpc73elq8og
Date	2026-02-25 12:45 EST
Device	Simulator/QPU

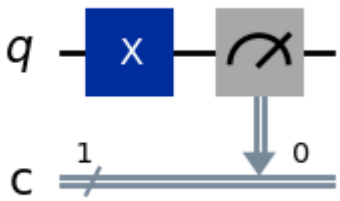
2.1 Quantum Circuit Diagram

In [2]:
```
import qiskit
import matplotlib

circuit = qiskit.QuantumCircuit(1, 1)
circuit.x(0)
circuit.measure(0, 0)

circuit.draw(output = 'mpl')
```

Out[2]:



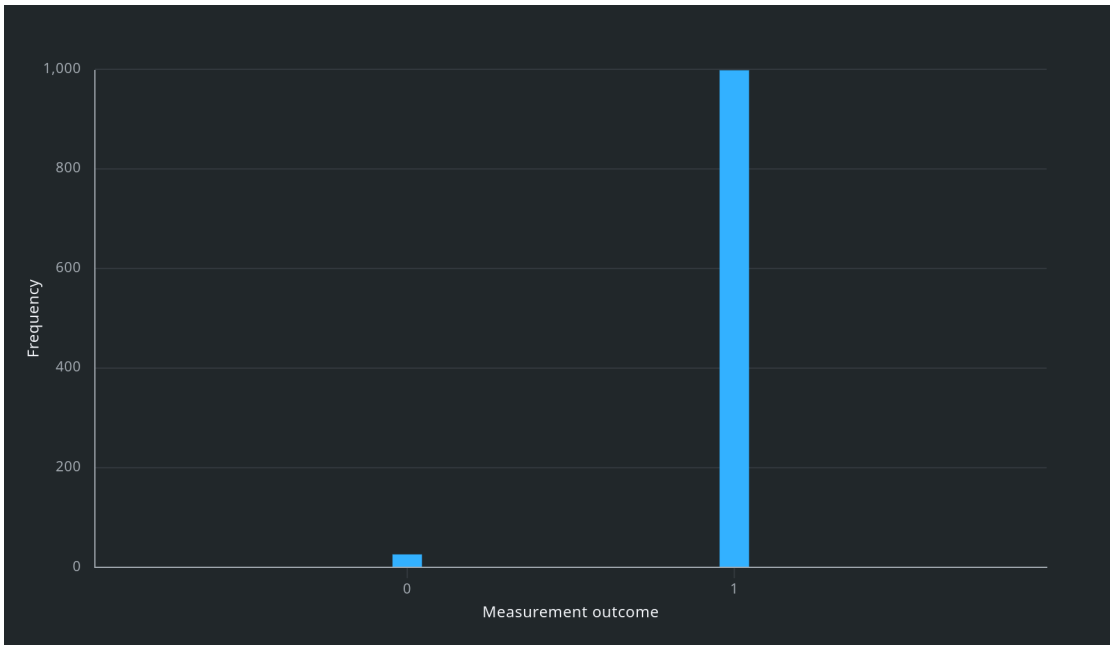
2.2 QASM Code|

In [ ]:
```
OPENQASM 2.0;
include "qelib1.inc";
qreg q[1];
creg c[1];
x q[0];
measure q[0] -> c[0];
```

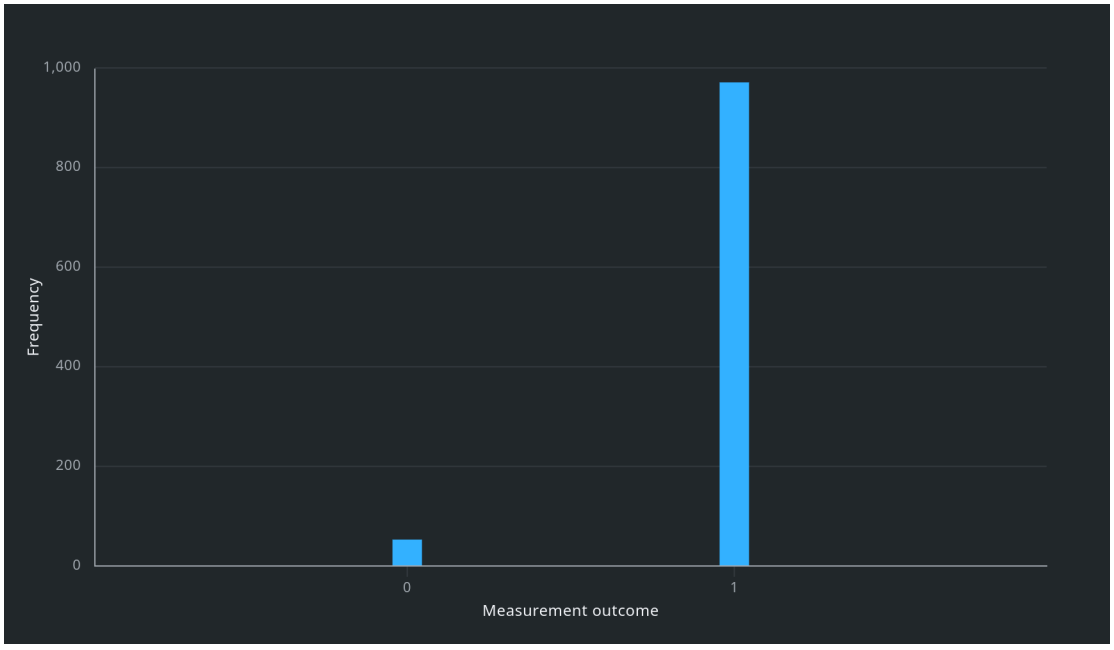
3. Methods

3.1 IBMQ Histogram Screenshot

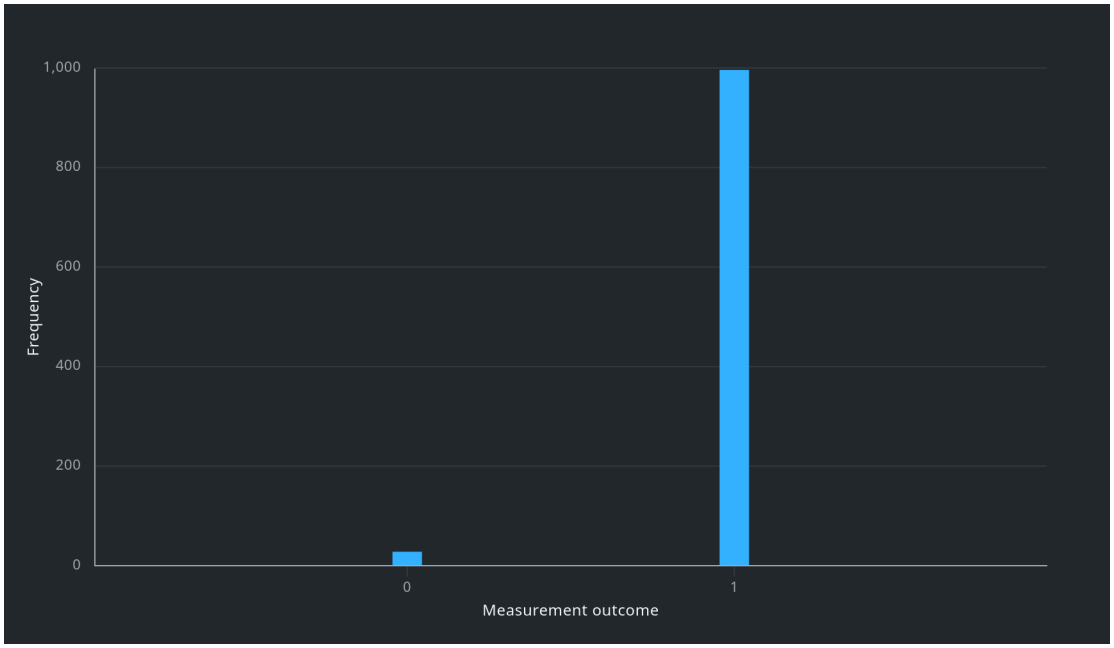
Run 1:



Run 2:



Run 3:



3.2 Raw Data Table

Run 1

State	Counts	Probability
|0⟩	26	2.54%
|1⟩	998	97.46%
Total	1024	100%

Run 2

State	Counts	Probability
|0⟩	53	5.18%
|1⟩	971	94.82%
Total	1024	100%

Run 3

State	Counts	Probability
|0⟩	28	2.73%
|1⟩	998	97.27%
Total	1024	100%

3.3 Python Simulation

In [ ]:
```
import numpy as np

shots = 1024

## Simulate Pauli-X Flip gate ##

Hgate_outcomes = np.random.choice(["|0>","|1>"], shots, p = [0,1])
kets, counts = np.unique(Hgate_outcomes, return_counts = True)
```

```
# Xgate_outcomes = np.random.choice(["|0>","|1>"], shots, p = [0,1])
# kets, counts = np.unique(Xgate_outcomes, return_counts = True)
print("==== Circuit 2 X-Gate ====")
print(f"|0> : {0 : .2%}, |1> : {1 : .2%}")
```

```
==== Circuit 2 X-Gate ====
|0> :  0.00%, |1> :  100.00%
```

4 Discussion

This data can be reproduced, by accessng the data from the Job Id stated in section 1. With 1024 shots, the error of the probabilities should be within ±3%. Although it was expected to be 100% probability to get the |1⟩ state, several factors such as quantum error and backend noise when applying the gate, lead to a small chance of remaining in the |0⟩ state.